

2021-Ph-H1-Q21

Section: Electricity

Topic: Current, PD, Power, Resistance

A resistor of resistance R has voltage V and current I . What is the power dissipated?

Solution:

Use:

$$P = IV = I^2 R = \frac{V^2}{R}$$

All three are valid — depends on known quantities.

Answer: E

Guidance for Students:

You should memorise all three power equations — use the one with values given.

Revision Tip:

Power equations:

$$P = IV, \quad P = I^2 R, \quad P = \frac{V^2}{R}$$

Choose based on what values you have.